Algebra and Trigonometry **Name (Print):** ____________________
MATH 1450-01

- This exam contains 14 pages (including the front and back of this cover page) and 16 problems. Check to see if any pages are missing.
- This is the third exam and will count for 20% of your final grade. You have 120 minutes (5:00 pm - 7:00 pm) to complete this exam. You may use one index card of notes and a calculator on this exam, but no other outside sources may be consulted.

The following rules apply:

- Every problem on this exam, other than true-or-false problems, requires work of some form. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. **A correct answer that has no supporting work will not receive full credit.**
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- There are 90 points total in this exam, as well as 1 bonus questions worth a total of 3 possible points.

Page:	3	4	5	6	8	9	10	11	12	13	Total
Points:	12	10	10	15	10	5	10	5	5	8	90
Score:											

Bonus:	

1. Mark each statement as $\boxed{\text{True}}$ or $\boxed{\text{False}}$ by completely shading in the box corresponding to your answer.

$\boxed{2 \text{ pts}}$ (a) $\sec(x)\csc(x)\cot(x) = \csc^2(x)$.

$\boxed{\text{True}}$ $\boxed{\text{False}}$

$\boxed{2 \text{ pts}}$ (b) $\dfrac{\sec(x)}{\tan(x)} = \cos(x)$.

$\boxed{\text{True}}$ $\boxed{\text{False}}$

$\boxed{2 \text{ pts}}$ (c) $\tan\left(\dfrac{3\pi}{4}\right) = \dfrac{-1+\sqrt{3}}{1+\sqrt{3}}$

$\boxed{\text{True}}$ $\boxed{\text{False}}$

$\boxed{2 \text{ pts}}$ (d) The absolute value of the complex number $-5.6 - 6.8i$ is approximately 8.81.

$\boxed{\text{True}}$ $\boxed{\text{False}}$

$\boxed{2 \text{ pts}}$ (e) In polar form, $6i = 6\text{cis}\left(\dfrac{\pi}{2}\right)$

$\boxed{\text{True}}$ $\boxed{\text{False}}$

$\boxed{2 \text{ pts}}$ (f) $\dfrac{\cot(\theta)}{\tan(\theta)} = \csc^2(\theta)\sec^2(\theta)$

$\boxed{\text{True}}$ $\boxed{\text{False}}$

5 pts 2. Simplify the expression

$$\frac{\tan\left(\frac{4}{7}x\right) + \tan\left(\frac{3}{7}x\right)}{1 - \tan\left(\frac{4}{7}x\right)\tan\left(\frac{3}{7}x\right)}$$

(A) $\tan(x)$ (B) $\tan(7x)$ (C) $\tan(3x)$ (D) $\tan(-4x)$

5 pts 3. Find the exact value of $\tan\left(\dfrac{13\pi}{12}\right)$.

(A) -1 (B) $\dfrac{1}{2}$ (C) 1 **(D)** $2 - \sqrt{3}$

5 pts 4. Choose the option below that is equal to $4\sin(7x)\cos(7x)$.

(A) $2\sin(14x)$ (B) $2\cos(14x)$ (C) $\sin\left(\frac{7}{2}x\right)$ (D) $\sin(3x) - \sin(4x)$

5 pts 5. Choose the option below that is equal to $10\sin(9x)\sin(5x)$.

(A) $5\left[\sin(14x) - \cos(4x)\right]$

(B) $5\left[\cos(4x) - \cos(14x)\right]$

(C) $10\cos(2x)\cos(7x)$

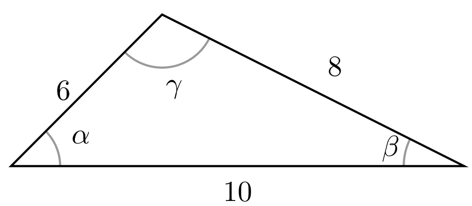
(D) $20\sin(7x)\cos(2x)$

5 pts 6. Simplify the expression to one term:

$$\sin(x)\cos(10x) - \cos(x)\sin(10x).$$

(A) $\tan\left(\frac{x}{10}\right)$ (B) $\cos(11x)$ **(C)** $-\sin(9x)$ (D) $\cos(9x)$

10 pts 7. Find the area of the triangle below. Round your answer to the nearest tenth.



(A) 31.9 (B) 281.3 (C) 303.6 **(D)** 24

8. Solve the equation exactly on the interval $0 \leq \theta < 2\pi$.

$$\frac{\sin(2\theta)}{\sqrt{3}\sin(\theta)} = 1$$

(A) $\theta = \frac{\pi}{12}$, $\theta = \frac{11\pi}{12}$, $\theta = \frac{13\pi}{12}$, and $\theta = \frac{23\pi}{12}$

(B) $\boldsymbol{\theta = \frac{\pi}{6}}$, $\boldsymbol{\theta = \frac{11\pi}{6}}$, $\boldsymbol{\theta = \frac{13\pi}{6}}$, and $\boldsymbol{\theta = \frac{23\pi}{6}}$

(C) $\theta = \frac{\pi}{24}$, $\theta = \frac{11\pi}{24}$, $\theta = \frac{13\pi}{24}$, and $\theta = \frac{23\pi}{24}$

(D) $\theta = \frac{\pi}{3}$, $\theta = \frac{11\pi}{3}$, $\theta = \frac{13\pi}{3}$, and $\theta = \frac{23\pi}{3}$

10 pts 9. Solve the equation exactly on the interval $[0, 2\pi)$.

$$\sin^2(x)(1-\cos^2(x)) - \cos^2(x)(1-\sin^2(x)) = 0$$

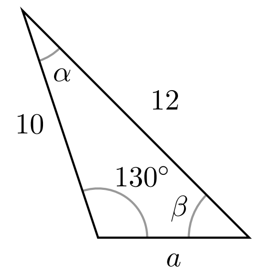
(A) $x = \frac{\pi}{12}$, $x = \frac{\pi}{4}$, $x = \frac{5\pi}{12}$, and $x = \frac{7\pi}{12}$.

(B) $x = \frac{\pi}{8}$, $x = \frac{3\pi}{8}$, $x = \frac{5\pi}{8}$, and $x = \frac{7\pi}{8}$.

(C) $x = \frac{\pi}{4}$, $x = \frac{3\pi}{4}$, $x = \frac{5\pi}{4}$, **and** $x = \frac{7\pi}{4}$.

(D) $x = \frac{\pi}{2}$, $x = \frac{3\pi}{2}$, $x = \frac{5\pi}{2}$, and $x = \frac{7\pi}{2}$.

5 pts 10. Solve for a in the triangle below. Round your answer to the nearest tenth.



(A) 2.2 (B) 4.1 **(C)** 2.8 (D) 3.1

5 pts 11. Convert the point $(2\sqrt{2}, 2\sqrt{2})$ from rectangular coordinates into polar coordinates (r, θ) with $0 \le r$ and $0 \le \theta < 2\pi$.

(A) $\left(2\sqrt{2}, \frac{3\pi}{4}\right)$ (B) $\left(8, \frac{\pi}{4}\right)$ **(C)** $\left(4, \frac{\pi}{4}\right)$ (D) $\left(4, \frac{5\pi}{4}\right)$

5 pts 12. Let z_1 and z_2 be complex numbers given in polar form by

$$z_1 = 25\text{cis}\left(\frac{13\pi}{12}\right), \text{ and } z_2 = 5\text{cis}\left(\frac{\pi}{12}\right).$$

Find $\frac{z_1}{z_2}$ in polar form.

(A) $5\textbf{cis}\left(\frac{3\pi}{4}\right)$ (B) $\sqrt{5}\text{cis}\left(\frac{3\pi}{4}\right)$ (C) $5\text{cis}\left(\frac{7\pi}{6}\right)$ (D) $\frac{1}{5}\text{cis}\left(\frac{3\pi}{4}\right)$

$\boxed{5 \text{ pts}}$ 13. Solve the system by any method.

$$\begin{cases} 5x - 5y &= 5/4 \\ 4x - 5y &= 1 \end{cases}$$

$x =$ ____________________, $y =$ ____________________

5 pts 14. The admission fee at a small fair is \$1.50 for children and \$4.00 for adults. On a certain day, 2200 people enter the fair and \$5050 is collected. How many children and how many adults attended?

Adults:__________________, Children:__________________

8 pts 15. Solve the system by any method.

$$\begin{cases} x - 2y + 3z & = 7 \\ 2x + y + z & = 4 \\ -3x + 2y - 2z & = -10 \end{cases}$$

$x =$ ________________, $y =$ ________________ $z =$ ________________

3 (bonus) 16. Solve the system by any method.

$$\begin{cases} y &= x^2 - 2x + 8 \\ y &= 5x - 2 \end{cases}$$